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NOMINAL SHOCKS, ENDOGENOUS GROWTH AND THE BUSINESS CYCLE*

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The paper proposes a simple model of wage setting and imperfect competition that takes into account knowledge and human capital accumulation. We show that, given increasing returns to reproducible factors, transitory disturbances to output that originate on the demand side of the economy produce permanent upward shifts in the aggregate production function. This implies that the presence of a stochastic trend in the process for income may not be informative *per se* about the forces driving the cycle.

In this paper we consider the effects of monetary shocks in a business cycle model with nominal rigidities, assuming increasing returns in the production of accumulable factors. As is well known, increasing returns in human and/or physical capital have represented the key hypothesis in the recent wave of growth models. These, however, assume perfectly flexible prices, thus ruling out the possibility of disequilibrium unemployment, on which we now focus. A related question we ask is whether this kind of analysis can cast any light on empirical evidence from, for instance, Nelson and Plosser (1982) and Campbell and Mankiw (1987), that the trend component of an economy's GNP includes a substantial random element.[1] These findings have undermined traditional views of the business cycles according to which fluctuations are primarily driven by nominal disturbances that do not alter the long-run performance of the economy. Real business cycle models, which are based on the neoclassical growth model, generate a stochastic trend in output if the process governing the Solow residual contains a unit root.[2] It has been pointed out that transitory technology or fiscal shocks will have permanent effects in endogenous growth models (see King *et al.* (1988) and Bean (1990)). Our aim here is to extend the analysis to nominal shocks. Path-breaking papers in this direction are Stadler (1986, 1990). He shows using a model of wage setting the possibility that in the presence of 'learning by doing' in production there is long-run non-neutrality of money. Since he works in a perfect competition framework he has to treat knowledge as a pure public good, i.e. not only non-rival but also totally non-excludable. Moreover the assumptions he makes about technology imply an explosive (or implosive) process for *per capita* income. The specification of technology proposed below, which is closer to the conventions of the new growth theory, avoids these shortfalls and makes clear the importance of constant returns in capital accumulation as a propagation factor due to which

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[1] See also Christiano and Eichenbaum (1990), who stress the difficulty of providing a compelling case that real GNP is either trend or difference stationary.

[2] See, for instance, Prescott (1986).

transitory nominal disturbances may have hysteretic effects. This shows that findings of a unit root in the process for income may say very little about the forces driving the cycle. It also raises doubts about the Blanchard and Quah (1989) approach to the decomposition of income between a stochastic trend and a stochastic cycle, based on the assumption that demand disturbances have no long-run effects on income.

I. THE MODEL

We formalise our argument by means of a simple model of wage setting and imperfect competition. Firms and workers are assumed to know the structure of the model and the distribution of the innovation terms that enter the model and to form expectations that fully reflect that knowledge. The aggregate economy is composed of n monopolistically competitive firms. Uncertainty enters in the form of aggregate demand disturbances that affect all industries identically. Each firm maximises its stream of discounted expected profits:

$$V_t = \max_{\{L_t^i, K_t^i\}} \mathrm{E}\left[\sum_{j=0}^{\infty} \theta^j \left(P_{t+j}^i Y_{t+j}^i - W_{t+j}^i L_{t+j}^i - r_{t+j} K_{t+j}^i\right) \mid I_t\right], \tag{1}$$

where θ is a time discount factor, P_t^i the price at which the firm i sells its product Y_t^i, L_t^i is employment by firm i, hired at the rental rate W_t^i, and r_t is the rental rate of capital K_t. The notation $\mathrm{E}[\cdot \mid I_t]$ denotes the expectation conditional on information available at time t, I_t. At time t, the demand facing each firm is given by:

$$y_t^i = -\omega(p_t^i - p_t) + (m_t - p_t). \tag{2}$$

Here and in what follows lower case letters indicate logarithms. m_t is the logarithm of the stock of nominal money and p_t the logarithm of the general price index defined as:

$$P_t \equiv n^{\frac{1}{\omega-1}}\left[\sum_{i=1}^{n} (P^i)^{1-\omega}\right]^{\frac{1}{1-\omega}}. \tag{3}$$

ω is the constant elasticity of substitution between goods; it goes to infinity in the perfect competition case. The demand for each good can be derived assuming a quantity theory of money or a Clower constraint and the same CES index for consumption and investment.[3]

The utility function of each household (whose number is normalised to be equal to the number of firms) depends on C defined as

$$C = n^{\frac{1}{1-\omega}}\left[\sum_{i=1}^{n} (C^i)^{\frac{\omega-1}{\omega}}\right]^{\frac{\omega}{\omega-1}}, \tag{4}$$

where we have suppressed the household's index, since all households are identical, and where C^i is the purchase of the ith product as a consumption good.

[3] See Blanchard and Kiyotaki (1987) and Kiyotaki (1988).

The effect of investment on the accumulation of physical capital K_t is given by:

$$K_t = K_{t-1} + I_{t-1}, \tag{5}$$

where

$$I = n^{\frac{1}{1-\omega}} \left[\sum_{i=1}^{n} (I^i)^{\frac{\omega-1}{\omega}} \right]^{\frac{\omega}{\omega-1}}, \tag{6}$$

and where I^i is the purchase of the ith product as an investment good.

Each firm operates with same Cobb–Douglas technology:

$$y_t^i = \mu k_t^i + \beta l_t^i + \psi k_t + \eta a_t, \tag{7}$$

where a is human capital, defined as labour quality, assumed to be equal for all workers, and the parameter ψ captures the external effects from average physical capital. The assumption of imperfect competition allows us to consider the case of increasing returns at the firm level, for instances $\mu \geqslant 1$. Romer (1986), working in a perfect competition framework, has revived the Arrovian hypothesis of 'learning by doing through investing' as a rationale for increasing returns to capital. Dasgupta and Stiglitz (1988) notice, however, that 'learning by doing' is consistent with perfect competition only on the implausible condition that not even a fraction of the accumulation of knowledge that goes with the accumulation of capital can be appropriated at the firm level. In fact, otherwise, assuming constant returns to scale as regards the rival factors, which is a consequence of the principle of replication, if factors were paid their marginal productivity, profits would be negative. This makes it advisable to consider imperfect competition instead.[4]

The absence of costs of adjustment in the use of factors makes it possible to treat the firm's problem like a one-period problem. From profit maximisation and ignoring constants, here and in what follows, the following expression for labour demand can be obtained:

$$l_t^{i,d} = \frac{-w_t^i + m_t/\omega + (1 - 1/\omega)(p_t + \mu k_t^i + \psi k_t + \eta a_t)}{1 - \beta(1 - 1/w)}. \tag{8}$$

Each firm is in a bargaining relationship with $\underline{L}$ workers and $\underline{L}$ is assumed fixed. The nominal wage is set one period in advance. The full information level of employment is assumed to be fixed at the level L^*. Workers want to minimise the mean squared deviation of the logarithm of employment from its target level under the constraint represented by (8). Since the problem is a linear quadratic one its solution exhibits the certainty equivalence property, i.e. the nominal wage is set at the level implying the expected employment $\mathrm{E}(L_t \mid I_{t-1})$ is equal to L^*. So eliminating expected employment in firm i from (8) we obtain:

$$w_t^i = \frac{\mathrm{E}(m_t \mid I_{t-1})}{\omega} + (1 - 1/\omega)\left[\mathrm{E}(p_t \mid I_{t-1}) + \mu k_t^i + \psi k_t + \eta a_t\right] - \left[1 - \beta(1 - 1/\omega)\right] l^*. \tag{9}$$

Assuming equality between individual and average stocks of physical capital

[4] The condition for the non-negativity of equilibrium profits is $\mu \leqslant [\omega/(\omega - 1)] - \beta$.

we then have $w_t^i = w_t$, $l_t^{i,d} = l_t^d$. Substituting the expression for l_t^i from (8) in (7) the resulting expression for y_t^i in (2) and solving for p_t^i one gets:

$$p_t^i = \frac{[(1-1/\omega)\,p_t + m_t/\omega]\,(1-\beta) + (1/\omega)\,[\beta w_t - (\mu+\psi)\,k_t - \eta a_t]}{1-\beta(1-1/\omega)}, \qquad (10)$$

which makes clear that $p_t = p_t^i$, so that:

$$p_t = (1-\beta)\,m_t + \beta w_t - (\mu+\psi)\,k_t - \eta a_t. \qquad (11)$$

From this it can be seen that the real cost of capital will be the same for all firms so that the assumption of each firm hiring the same amount of capital is warranted. From (8) and (11), we then get:

$$l_t^d = m_t - w_t. \qquad (12)$$

From (12) and the wage rule one infers that:

$$l_t = m_t - \mathrm{E}(m_t \mid I_{t-1}) + l^*. \qquad (13)$$

Finally, the money supply is assumed to follow a random walk with positive drift π. The drift is set by the central bank so as to avoid an ongoing deflation in a growing economy, given a constant velocity of money. The money supply rule is public knowledge. This is:

$$m_t = m_{t-1} + \pi + \epsilon_t, \qquad (14)$$

where the disturbance term ϵ is a zero-mean stochastic error with constant variance.

Substituting the expression for l_t given by (13) in the production function, after calculating the forecast error from (14), the following process for income is derived:

$$y_t = \beta(\epsilon_t + l^*) + (\mu+\psi)\,k_t + \eta a_t. \qquad (15)$$

So output depends on the level of physical and human capital and on technology, as well as on the full information level of employment. A positive value of ϵ_t raises output as in all demand-side models of the business cycle with nominal wage contracting because it causes the price level to rise above its expected value and hence depresses the real wage, increasing labour demand.

II. LONG-RUN GROWTH

We now come to the accumulation of factors of production. Stadler (1990) endogenises the growth process, introducing 'learning by doing' not appropriable at the firm level in a perfect competition wage setting monetary model of the business cycle. He posits for each firm i the production technology $Y_t^i = (L_t^i)^\alpha Z_t^{1-\alpha}$, $0 < \alpha < 1$, where Z_t is technical knowledge, assumed to evolve according to:

$$Z_t = Z_{t-1}\left(\frac{Y_{t-1}}{L_{t-1}}\right)^\lambda (L_{t-1})^\gamma \quad \text{with} \quad 0 < \lambda, 0 < \gamma < 1.$$

The accumulation law of this economy is then $Z_t/Z_{t-1} = Z_{t-1}^{\lambda(1-\alpha)} L_{t-1}^{\lambda(\alpha-1)+\gamma}$. A problem with this formulation is that it may imply explosive growth. The following alternative avoids this and makes it easier to relate the analysis to the literature on growth. As regards the accumulation of physical capital, we make the Solovian behavioural assumption that the ratio of investment to income, s, is constant. We then have:

$$\frac{K_t - K_{t-1}}{K_{t-1}} = sL_{t-1}^{\beta} A_{t-1}^{\eta} K_{t-1}^{\mu+\psi-1} - \delta_2. \tag{16}$$

For human capital we consider a transition equation based on a Cobb–Douglas technology:

$$\frac{A_t^i - A_{t-1}^i}{A_{t-1}^i} = (L_{t-1}^i)^{\sigma} L_{t-1}^{\zeta} (A_{t-1}^i)^{\vartheta-1} A_{t-1}^{\xi} (K_{t-1}^i)^{\chi} K_{t-1}^{v} - \delta_1, \tag{17}$$

where δ_1 is the rate of depreciation, and ζ, ξ, v represent spill-over effects. While in Lucas (1988) human capital is accumulated through formal schooling, here a different effect is at work, i.e. that having worked in the past makes workers more productive, as happens when there is 'on the job training'. Notice that, in offering labour, workers would in general take into account the increase in their human capital that being involved in production brings about, abstracting from external effects on labour.[5] By symmetry, (17) can be written as:

$$\frac{A_t - A_{t-1}}{A_{t-1}} = L_{t-1}^{\zeta+\sigma} A_{t-1}^{\xi+\vartheta-1} K_{t-1}^{v+\chi} - \delta_1. \tag{18}$$

In steady state, monetary shocks do not impinge on the economy, as expectations are realised, so $L_{ss} = L^*$. As can be seen by second-order differencing of a_t and k_t, for A and K to grow at a constant rate we need:

$$0 = s(L^*)^{\beta} A_{t-1}^{\eta} K_{t-1}^{\mu+\psi-1}\left[1 - \frac{(1+g_k)^{1-\mu-\psi}}{(1+g_a)^{\eta}}\right], \tag{19}$$

$$0 = (L^*)^{\zeta+\sigma} A_{t-1}^{\xi+\vartheta-1} K_{t-1}^{v+\chi}\left[1 - \frac{(1+g_a)^{1-\xi-\vartheta}}{(1+g_k)^{v+\chi}}\right]. \tag{20}$$

If $g_k = g_a = g$, the rate of balanced growth, is to be strictly positive, it must be that: $\mu+\psi+\eta = 1$, $\xi+\vartheta+\chi+v = 1$. This is the usual condition of constant returns to scale to the producible factors in their own production. If $\delta_1 = \delta_2 = \delta$ it is possible to get a closed form for the steady-state rate of growth. This is:

$$g = -\delta + (L^*)^{\frac{\beta(v+\chi)+(\zeta+\sigma)\eta}{v+\chi+\eta}} s^{\frac{v+\chi}{v+\chi+\eta}}. \tag{21}$$

[5] In fact, we had assumed a unified labour market, equilibrium employment would have been higher through this channel.

This shows that the rate of saving and the equilibrium level of employment are positively related with the rate of growth, while the rate of depreciation is negatively related to it.[6]

III. FROM THE SHORT TO THE LONG RUN

To study the transitional dynamics we will have to resort to some simplifying assumptions, in fact considering two different models nested in the general one used for steady-state analysis.

In the first model human capital does not enter the production function, i.e. $\eta = 0 = \sigma$. Now consider the hypothesis that the 'learning by doing' effect is not so strong as to deliver constant returns to scale to the accumulable factor. This implies that $\mu + \psi < 1$. The steady-state level of physical capital is:

$$K_{ss}^{1} = \left[\frac{s(L^{*})^{\beta}}{\delta_2}\right]^{\frac{1}{1-\psi-\mu}}. \tag{22}$$

We will show that given these assumptions on technology, an innovation in the process governing money supply cannot have permanent effects. Suppose, for instance, that, starting from a situation of steady-state equilibrium, we have at time t a positive forecast error, so that $L_t > L^{*}$. To isolate the effect of this single shock, let us assume that in all subsequent periods expectations are realised, i.e. $L_s = L^{*}$ for all $s > t$. But then from (16) and (22) we see that there will be a positive net investment, i.e. $K_{t+1} > K_t = K_{ss}$. However, subsequently capital will decumulate, as the same equations imply for $K_s > K_{ss}$. The reduction in the stock of capital will go on until the stationary-state level is reached again.

Now consider instead the case of constant returns. Equation (16) can then be rewritten, as an approximation:

$$k_t = k_{t-1} + sL_{t-1}^{\beta} - \delta_2, \tag{23}$$

but since from (7) we have:

$$k_{t-1} = y_{t-1} - \beta l_{t-1}, \tag{24}$$

we obtain, substituting in (15), the following data generating process for income:

$$y_t = y_{t-1} - \delta_2 + \beta(\epsilon_t - \epsilon_{t-1}) + s(L^{*})^{\beta} \exp \beta\epsilon_{t-1}. \tag{25}$$

The process is integrated of order one and the structure of the error term is such that shocks will have permanent effects. This is clear by considering the impulse response function:

$$y_t = y_0 - \delta_2 t + \beta(\epsilon_t - \epsilon_0) + s(L^{*})^{\beta} \sum_{i=0}^{t-1} \exp \beta\epsilon_{t-i}. \tag{26}$$

In the second particular case physical capital is ignored, as in Stadler (1990),

[6] This is true even with different rates of depreciation as can be seen by considering the following equation in g:

$$g + \delta_2 = [s(L^{*})^{\frac{\beta(\upsilon+\chi)+(\zeta+\sigma)(\eta)}{\upsilon+\chi}}]/(g+\delta_1)^{\frac{\eta}{\upsilon+\chi}}.$$

We have on the left-hand side of the equation a linear function of g and on the right-hand side a hyperbola. By analysing how the intersection between the two curves moves when the values of the parameters change the conclusion follows.

i.e. $\mu + \psi = 0$ and $\chi + v = 0$. If $\xi + \vartheta < 1$, the steady-state level of human capital will be:

$$A^2_{ss} = \left[\frac{(L^*)^{\sigma+\zeta}}{\delta_1}\right]^{\frac{1}{1-\xi-\vartheta}}. \tag{27}$$

If we start with this stock of human capital any shock raising employment above the full information level will push the stock above the steady-state value. However, this over-accumulation will subsequently be undone, unless further shocks occur in the same direction of the first.

Again for long-run growth we need constant returns to scale, i.e. $\omega + \vartheta = 1$. Through manipulations analogous to those just described we again get a unit root process for income:

$$y_t = y_{t-1} - \delta_1 + \beta(\epsilon_t - \epsilon_{t-1}) + (L^*)^{\zeta+\sigma} \exp(\zeta+\sigma)\,\epsilon_{t-1} \tag{28}$$

and
$$y_t = y_0 - \delta_1 t + \beta(\epsilon_t - \epsilon_0) + (L^*)^{\zeta+\sigma} \sum_{i=0}^{t-1} (\zeta+\sigma) \exp \beta\epsilon_{t-i}. \tag{29}$$

So it is clear from the analysis of both models that 'learning by doing', when it is not strong enough to imply increasing returns, can deliver persistence not hysteresis. The following assertion by Stadler (1990) is therefore somewhat too general. He writes: 'when technology is dependent on demand conditions, monetary shocks have a permanent impact on output. There is a long-run non-neutrality of money in models with endogenous technology'. However we have shown that endogeneity *per se* is not sufficient to get the result.

On the other hand, when the condition for unbounded accumulation holds, i.e. the presence of non-decreasing returns to the producible factors, nominal demand disturbances will have permanent hysteretic effects. In fact output is a difference-stationary process that depends upon the sum of all past monetary misperceptions.

In our approach to the study of the interactions between growth and business cycles, recessions have negative long-run effects on productivity. Some recent theoretical papers have instead taken up the 'Schumpeterian' idea of recessions providing a mechanism for reducing resource misallocations, with a long-run positive effect on productivity. See among others Aghion and Saint-Paul (1991) and Caballero and Hammour (1994). We wish to underline that the two views, the 'Kaldorian' and the 'Schumpeterian', are not incompatible in the sense that a slump may have both a positive influence on productivity through reorganisation efforts, cleansing effects etc., and a negative influence, through reduced 'learning by doing' and 'on the job training'. The question of which kind of effects prevail is in fact an empirical one and the evidence so far is mixed. Gali and Hammour (1993) and Saint-Paul (1993) using economy-wide data (respectively for the United States and for 22 OECD countries) have found some evidence that the long-run response of productivity to demand shocks is negative. Jiménez and Marchetti (1995), using a panel data set of 402 four-digit US manufacturing industries find, on the contrary, that the response is positive or positively close to zero depending on the industry. We can conclude that both views therefore deserve further investigation theoretically and empirically.

IV. CONCLUDING REMARKS

The possibility that demand shocks have permanent effects contradicts a long tradition in macroeconomic theory: fluctuations in economic activity are generally studied in a 'short-term' framework in which the capital stock and productivity are kept constant. Conversely the supply of capital and labour plays the central role in understanding growth, but inflation and short term demand disturbances are ignored. The above separation is removed in Stadler (1986, 1990). The present paper has proposed a monetary, imperfect competition model of the business cycle, in which knowledge accumulation internal and external to firms is taken into account. The specification of technology avoids some limitations and counterfactual implications of Stadler's models, regarding in particular the appropriability of knowledge and the explosive character of growth. Furthermore, being in the mould of most endogenous growth models, it makes it easier to clarify the link between the conditions for endogenous growth and hysteresis in *per capita* income.

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